SCIENCE-2 END SEM-ANSWER KEY

Section A: MCQs

Q1. The pH of lemon juice is about 2.0 while tomato juice is about 4.0. What is the difference in hydrogen ion concentration?

Answer: d. **100 times**

Q2. A diet extremely low in fats might result in deficiency of:

Answer: c. **Steroid hormones**

Q3. "All birds I have seen have feathers…" This is:

Answer: b. **Inductive reasoning**

Q4. If hydrogen bonds between water molecules were stronger, perspiration cooling:

Answer: d. **Cooling becomes more difficult (requires more energy)**

Q5. Single amino acid substitution in protein primary structure results in:

Answer: c. **Tertiary structure may change, affecting function**

Q6. Drug prevents RNA synthesis → which organelle affected?

Answer: c. **Ribosome**

Q7. Plant cell plasmolysis occurs in which solution?

Answer: c. **Hypertonic**

Q8. Why is Carbon-14 used in dating fossils?

Answer: c. **Unstable isotope that decays at predictable rate**

Q9. Evolution as a theory means:

Answer: b. **Widely accepted, well-tested explanation**

Q10. Phospholipids are major components of:

Answer: a. **Plasma membrane**

Q11. If lysosomes rupture:

Answer: d. **Cell breaks down due to enzymes**

Q12. Endosymbiotic theory explains origin of:

Answer: b. **Mitochondria and chloroplasts**

Q13. Role of oxygen in ETC:

Answer: b. **Final electron acceptor**

Q14. No NAD+ in cell →

Answer: b. **Glycolysis stops, fermentation starts**

Q15. ATP used ("battery discharged"):

Answer: a. **Phosphate removed releasing energy**

Q16. Aerobic vs fermentation:

Answer: b. **Aerobic requires oxygen, fermentation does not**

Q17. Purpose of G2 checkpoint:

Answer: b. **Ensure DNA replication is complete and undamaged**

Q18. After Meiosis II, gametes are:

Answer: a. **Haploid**

Q19. Separation of sister chromatids occurs in:

Answer: b. **Anaphase II**

Q20. Blue × white flowers → light blue offspring:

Answer: c. **Incomplete dominance**

Q21. Diploid number 32 → daughter cells in mitosis:

Answer: c. **32**

Q22. Genes close on same chromosome:

Answer: b. **Linked, inherited together**

Q23. mRNA processing involves:

Answer: b. **5' cap + poly-A tail addition**

Q24. DNA in gel electrophoresis moves:

Answer: b. **Towards positive electrode**

Q25. Females prefer brightest males →

Answer: c. **Sexual selection**

Q26. Small group forms new population →

Answer: a. **Founder effect**

Q27. DNA has 28% Adenine → Cytosine?

Answer: a. **22%**

Q28. Reverse transcriptase shows:

Answer: c. **RNA → DNA possible**

Q29. Pelvic bones in whales:

Answer: b. **Vestigial structures**

Q30. Blocking A-site of ribosome:

Answer: b. **Protein elongation stops**

Section B: Short Answers

Q31. DNA vs RNA bases

Answer:

DNA contains **A, T, G, C**, while RNA contains **A, U, G, C**.

Difference: **Thymine (DNA) vs Uracil (RNA)**.

Uracil lacks a methyl group present in thymine → makes RNA less stable.

Q32. Glycolysis

Answer:

Glycolysis is the breakdown of glucose into **2 pyruvates** in cytoplasm.

- Investment phase: uses **2 ATP**
- Payoff phase: produces **4 ATP (net = 2 ATP)**
- Produces **2 NADH**

No oxygen is required.

Q33. Citric Acid Cycle

Answer:

Occurs in mitochondrial matrix.

Per glucose:

- **2 ATP (as GTP)**
- **6 NADH + 2 $FADH_2$**
- CO_2 released

These carriers go to ETC for ATP production.

Q34. Mendel's laws in meiosis

Answer:

- **Law of Segregation:** Anaphase I (alleles separate)
- **Law of Independent Assortment:** Metaphase I (random alignment of homologous chromosomes)

Q35. CRISPR Technology

Answer:

CRISPR-Cas9 is a gene-editing tool.

- Uses guide RNA to target DNA
- Cas9 enzyme cuts DNA
- Allows insertion/deletion of genes

Applications: medicine, agriculture, research.

Q36. RNA Splicing

Answer:

Removal of **introns** and joining of **exons** in pre-mRNA.

Occurs in nucleus using spliceosome → forms mature mRNA.

Q37. Human vs Banana genome complexity

Answer:

Humans achieve complexity via:

- **Alternative splicing**
- **Post-translational modifications**
- **Gene regulation networks**
- **Non-coding DNA roles**

Q38. Lac Operon

Answer:

Regulates lactose metabolism in bacteria.

- **Lactose present → operon ON**
- Repressor inactivated → transcription occurs
- Produces enzymes for lactose digestion.

Q39. Non-Mendelian inheritance

Answer:

Includes:

- Incomplete dominance
- Codominance
- Multiple alleles
- Polygenic traits
- Mitochondrial inheritance

Q40. Mechanisms of evolution

Answer:

- **Natural selection**
- **Genetic drift**
- **Mutation**
- **Gene flow**
- **Sexual selection**